REPORT

Title

Implementation of the Missionaries and Cannibals Problem Using Python

Aim

To implement the Missionaries and Cannibals problem in Python and determine a valid sequence of crossings that safely transports all missionaries and cannibals across the river.

Problem Statement

The Missionaries and Cannibals Problem is a river-crossing puzzle involving three missionaries and three cannibals. The objective is to move everyone from one side of the river to the other using a boat that can carry at most two people. Throughout the process, the number of cannibals must never exceed the number of missionaries on either river bank unless no missionaries are present. The program should find a valid sequence of crossings that satisfies all the given constraints.

Requirements

- Python 3.x

Procedure

1. Represent the number of missionaries, cannibals, and boat position as a state.
2. Define the initial and goal states.
3. Generate all possible boat movements.
4. Check whether each new state satisfies the problem constraints.
5. Ignore invalid and previously visited states.
6. Continue exploring valid states until the goal state is reached.
7. Display the sequence of valid crossings.

Program

```
from collections import deque


def is_valid(m, c):
    if m < 0 or c < 0 or m > 3 or c > 3:
        return False
    if (m > 0 and m < c):
        return False
    if ((3 - m) > 0 and (3 - m) < (3 - c)):
        return False
    return True


def bfs():
    start = (3, 3, 'L')
```

```
    goal = (0, 0, 'R')

    queue = deque([(start, [])])
    visited = set()

    moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        if state in visited:
            continue

        visited.add(state)

        m, c, boat = state

        for dm, dc in moves:
            if boat == 'L':
                new = (m - dm, c - dc, 'R')
```

```
            else:
                new = (m + dm, c + dc, 'L')

            if is_valid(new[0], new[1]):
                queue.append((new, path + [state]))

    return None

solution = bfs()

if solution:
    print("Solution Path:")
    for s in solution:
        print(s)
else:
    print("No Solution Found")
```

Output

```
Solution Path:

(3, 3, 'L')
(3, 1, 'R')
(3, 2, 'L')
```

```
(3, 0, 'R')
(3, 1, 'L')
(1, 1, 'R')
(2, 2, 'L')
(0, 2, 'R')
(0, 3, 'L')
(0, 1, 'R')
(1, 1, 'L')
(0, 0, 'R')
```

Result

The Missionaries and Cannibals problem was successfully implemented in Python. The program generated valid river crossings and safely transported all missionaries and cannibals to the opposite bank while satisfying all the constraints.

Conclusion

The Missionaries and Cannibals problem is a classic state-space search problem in Artificial Intelligence. It demonstrates the application of search techniques to constraint-based problems by systematically exploring valid states and eliminating invalid ones. The problem highlights the importance of state representation, constraint checking, and systematic exploration in finding feasible solutions.